

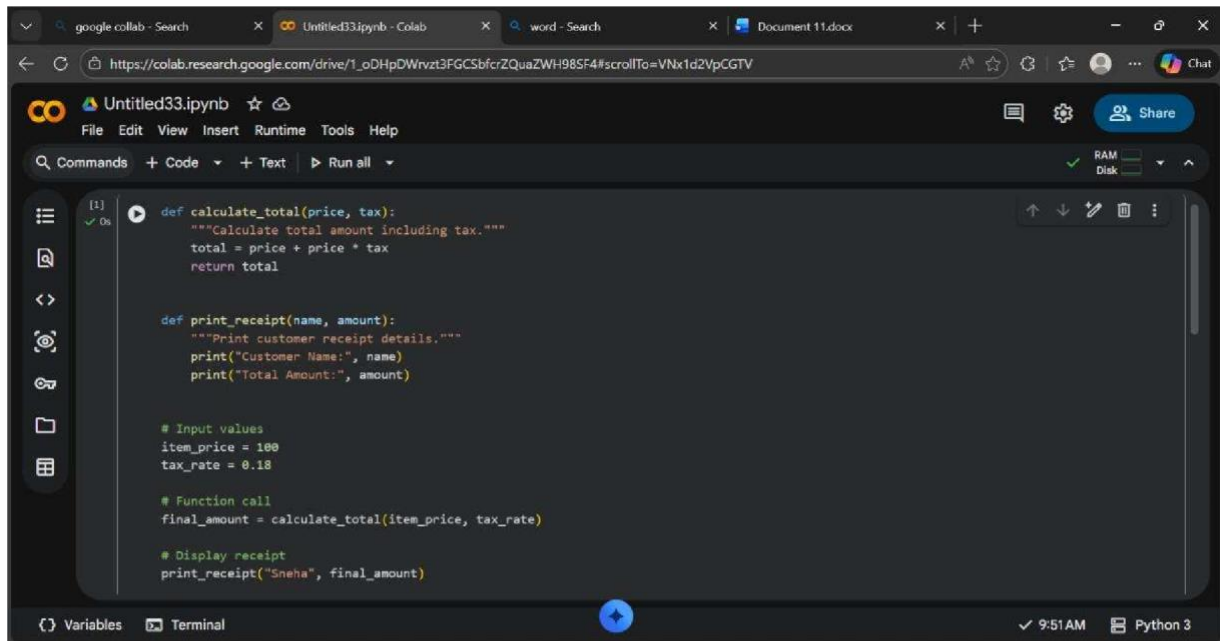
Assignment 10.4 Ai Assisted Coding

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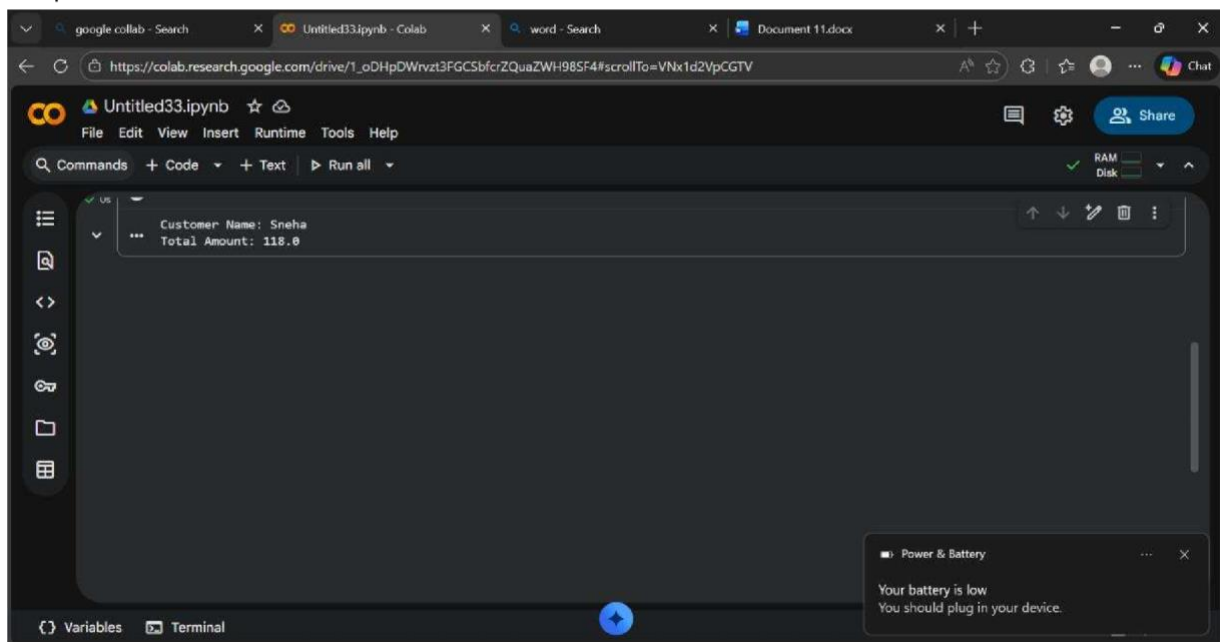
TASK 1:

Code:



```
[1] def calculate_total(price, tax):  
    """Calculate total amount including tax."""  
    total = price + price * tax  
    return total  
  
    def print_receipt(name, amount):  
        """Print customer receipt details."""  
        print("Customer Name:", name)  
        print("Total Amount:", amount)  
  
    # Input values  
    item_price = 100  
    tax_rate = 0.18  
  
    # Function call  
    final_amount = calculate_total(item_price, tax_rate)  
  
    # Display receipt  
    print_receipt("Sneha", final_amount)
```

Output:



```
Customer Name: Sneha  
Total Amount: 118.0
```

Explanation:

- >AI fixed syntax mistakes and indentation errors in the script.
- >It corrected wrong function calls and mismatched variable names.
- >Naming was standardized using proper Python conventions.
- >The code structure was cleaned and organized properly.
- >The final program runs correctly without errors.

Task 2:

Performance-Oriented Code Review

Scenario

A data processing function works correctly but is inefficient and slows down the system when large datasets are used.

Task Description

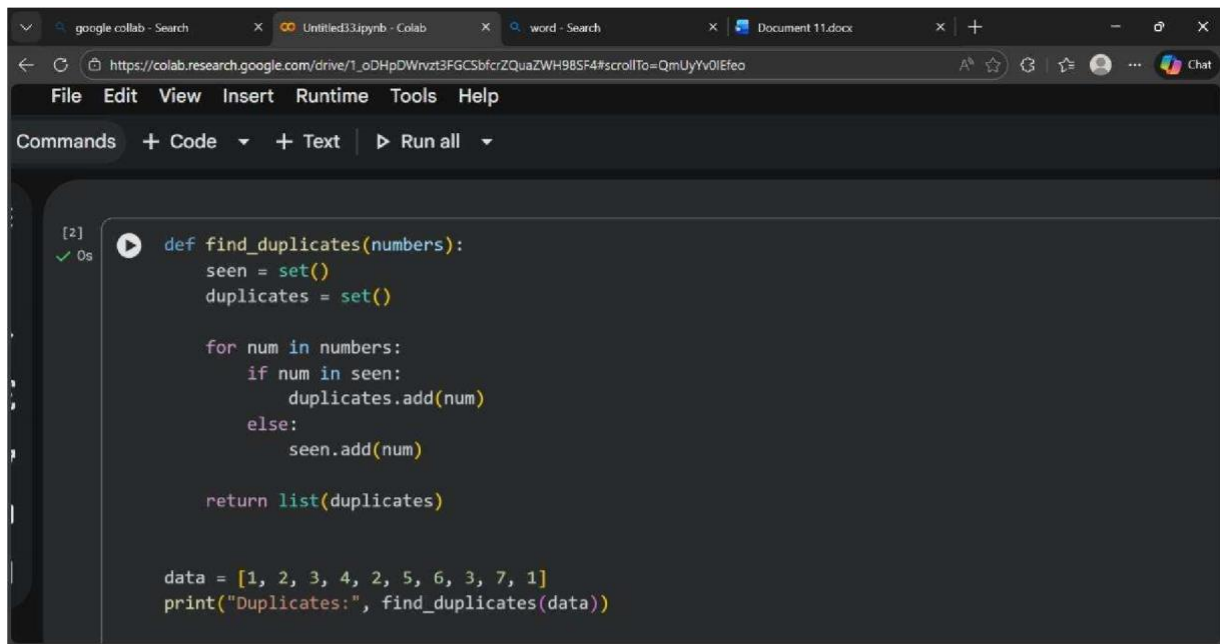
You are provided with a function that identifies duplicate values in a list using inefficient nested loops.

Using AI-assisted code review:

- Analyze the logic for performance bottlenecks
- Refactor the code for better time complexity
- Preserve the correctness of the output Ask the AI to explain:
 - Why the original approach was inefficient
 - How the optimized version improves performance

Expected Outcome

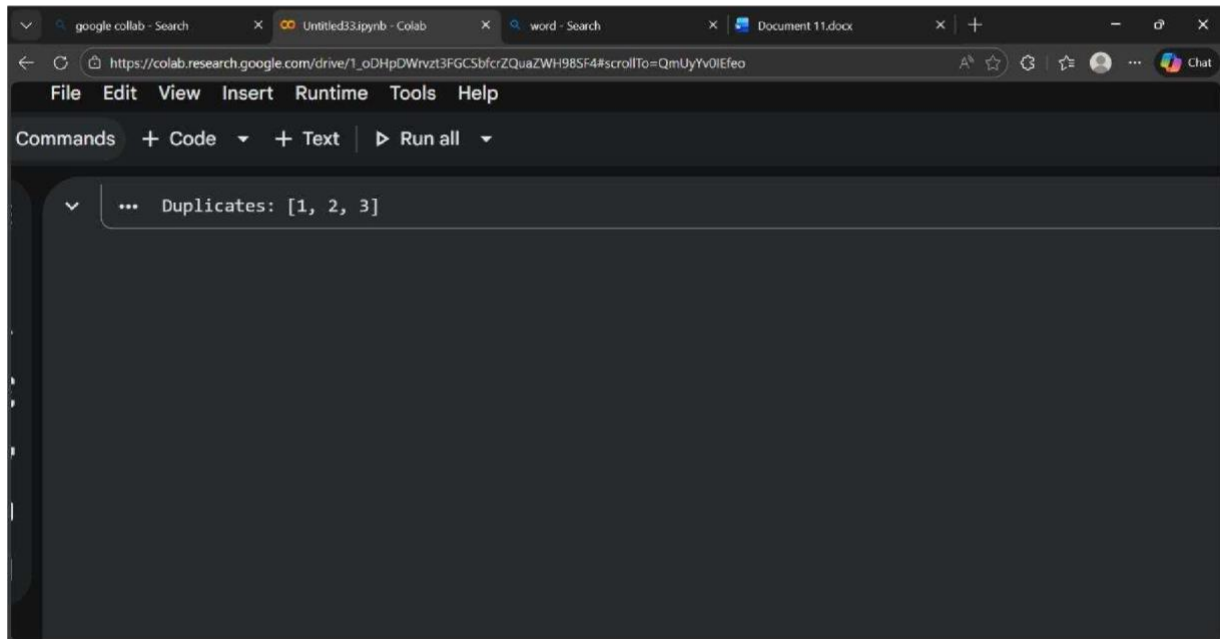
- Optimized duplicate-detection logic (e.g., using sets or hash-based structures)
- Improved time complexity
- AI explanation of performance improvement
- Clean, readable implementation Code:



The screenshot shows a Google Colab notebook interface. The top bar includes tabs for 'google collab - Search', 'Untitled33.ipynb - Colab', 'word - Search', and 'Document 11.docx'. The address bar shows the URL 'https://colab.research.google.com/drive/1_oDHpDWrvzt3FGCSbfc/ZQuaZWH98SF4#scrollTo=QmUyYv0IEfeo'. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu bar, there are tabs for 'Commands', '+ Code', '+ Text', and 'Run all'. The main code cell is labeled '[2]' and shows a Python function 'find_duplicates' that uses a set to find duplicates in a list. The function iterates through the list, adding elements to a 'seen' set and a 'duplicates' set. The output of the function is a list of duplicates. The code is as follows:

```
[2] def find_duplicates(numbers):  
    seen = set()  
    duplicates = set()  
  
    for num in numbers:  
        if num in seen:  
            duplicates.add(num)  
        else:  
            seen.add(num)  
  
    return list(duplicates)  
  
data = [1, 2, 3, 4, 2, 5, 6, 3, 7, 1]  
print("Duplicates:", find_duplicates(data))
```

Output:



The screenshot shows the same Google Colab notebook interface as the previous one, but now the output of the code cell is visible. The output is a list of duplicates: [1, 2, 3]. The output is displayed in a dark box with a light border, and the text 'Duplicates: [1, 2, 3]' is visible. The interface elements are the same as in the previous screenshot.

Explanation:

- >The original code used **nested loops**, comparing each element with every other element.
- >This caused **$O(n^2)$ time complexity**, making it slow for large lists.
- >The optimized version uses a **set** for quick lookup of seen elements.
- >Set operations work in **$O(1)$ time**, allowing duplicates to be found in one pass.

-->This reduces overall complexity to **$O(n)$** , improving performance while keeping correct results.

Task 3:

Readability and Maintainability Refactoring

Scenario

A working script exists in a project, but it is difficult to understand due to poor naming, formatting, and structure. The team wants it rewritten for long-term maintainability.

Task Description

You are given a poorly structured Python function with:

- Cryptic function names
- Poor indentation
- Unclear variable naming
- No documentation

Use AI-assisted review to:

- Refactor the code for clarity
- Apply PEP 8 formatting standards
- Improve naming conventions
- Add meaningful documentation

Expected Outcome

- Clean, well-structured code
- Descriptive function and variable names
- Proper indentation and formatting

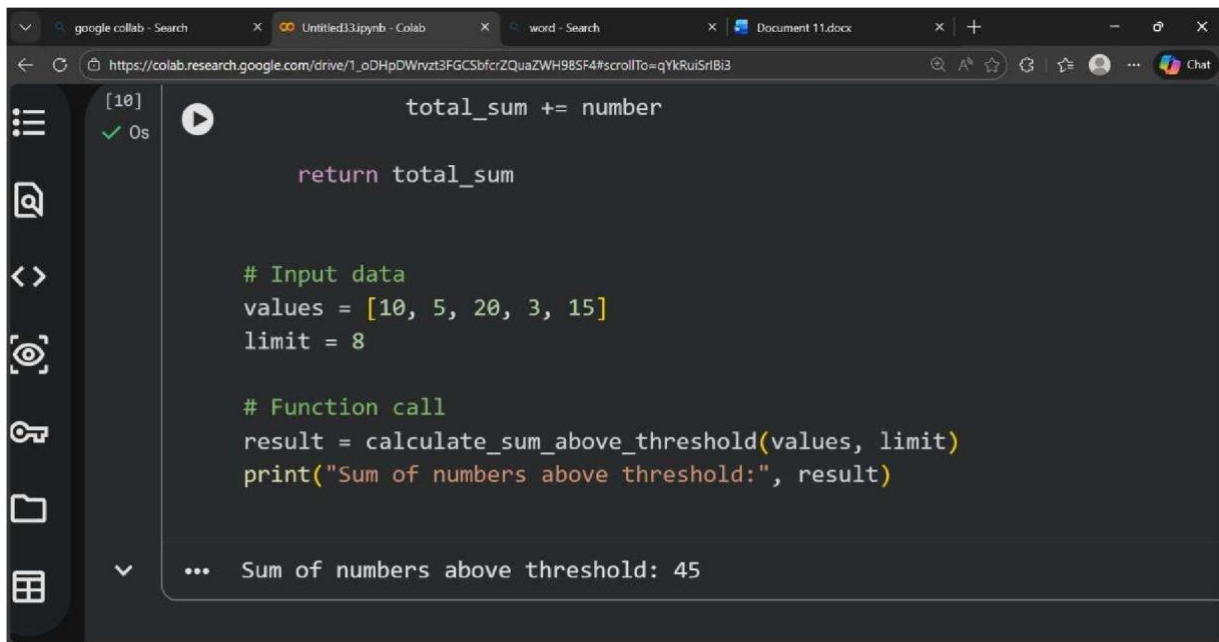
- Docstrings explaining the function purpose
- AI explanation of readability improvements Code:

The image displays two sequential screenshots of a Google Colab notebook titled 'Untitled33.ipynb'. The top screenshot shows the definition of a function `calculate_sum_above_threshold` with a docstring explaining its purpose and parameters. The bottom screenshot shows the same function being tested with sample data.

```
[10] def calculate_sum_above_threshold(numbers, threshold):  
    """  
    Calculate the sum of numbers greater than a given threshold.  
  
    Parameters:  
    numbers (list): List of numeric values.  
    threshold (int or float): The minimum value to include in the sum.  
  
    Returns:  
    int or float: Sum of numbers greater than the threshold.  
    """  
    total_sum = 0  
  
    for number in numbers:  
        if number > threshold:  
            total_sum += number  
  
    return total_sum
```

```
[10] int or float: Sum of numbers greater than the threshold.  
    """  
    total_sum = 0  
  
    for number in numbers:  
        if number > threshold:  
            total_sum += number  
  
    return total_sum  
  
    # Input data  
    values = [10, 5, 20, 3, 15]  
    limit = 8  
  
    # Function call  
    result = calculate_sum_above_threshold(values, limit)  
    print("Sum of numbers above threshold:", result)
```

Output:



The screenshot shows a Google Colab notebook interface. The top bar includes tabs for 'google colab - Search', 'Untitled33.ipynb - Colab', 'word - Search', and 'Document 11.docx'. The address bar shows a URL from colab.research.google.com. The notebook content area has a dark background and displays the following Python code:

```
[10] ✓ 0s
def calculate_sum_above_threshold(values, limit):
    """Calculate the sum of numbers above a threshold.

    Parameters:
        values (list): A list of integers.
        limit (int): A threshold value.

    Returns:
        int: The sum of numbers greater than the limit.
    """
    total_sum = 0
    for number in values:
        if number > limit:
            total_sum += number

    return total_sum

# Input data
values = [10, 5, 20, 3, 15]
limit = 8

# Function call
result = calculate_sum_above_threshold(values, limit)
print("Sum of numbers above threshold:", result)
```

Below the code, the output is displayed: `... Sum of numbers above threshold: 45`. The left sidebar contains icons for file management, search, and other Colab features.

Explanation:

-->The original code was hard to understand due to unclear function and variable names, poor formatting, and no documentation.

--> The refactored version improves readability by using a descriptive function name and meaningful variable names.

-->Proper indentation and spacing were applied following PEP 8 standards. A docstring was added to explain the function's purpose, parameters, and return value.

--> These changes make the code easier to read, maintain, and modify in the future.

Task 4:

Secure Coding and Reliability Review

Scenario

A backend function retrieves user data from a database but has security vulnerabilities and poor error handling, making it unsafe for production deployment.

Task Description

You are given a Python script that:

- Uses unsafe SQL query construction
 - Has no input validation
 - Lacks exception handling
- Use AI tools to:

- Identify security vulnerabilities
- Refactor the code using safe coding practices
- Add proper exception handling
- Improve robustness and reliability

Expected Outcome

- Secure SQL queries using parameterized statements
 - Input validation logic
 - Try-except blocks for runtime safety
 - AI-generated explanation of security improvements
 - Production-ready code structure
- Code:

The screenshot shows the Google Colab interface for a notebook titled 'Untitled33.ipynb'. The code editor contains the following Python code:

```
[29] import sqlite3

def get_user(username):
    """
    Retrieve user details safely from the database.

    Parameters:
    username (str): Username entered by the user.

    Returns:
    tuple or None: User record if found, otherwise None.
    """
    # Input validation
    if not username.isalnum():
        print("Invalid username. Only letters and numbers allowed.")
        return None
```

The interface includes a top navigation bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help' menus. A 'Share' button is in the top right. Below the code editor is a 'Variables' and 'Terminal' section. The status bar at the bottom shows '10:26 AM' and 'Python 3'.

The screenshot shows the continuation of the Python script in the Google Colab interface. The code editor contains the following Python code:

```
        return None

    try:
        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()

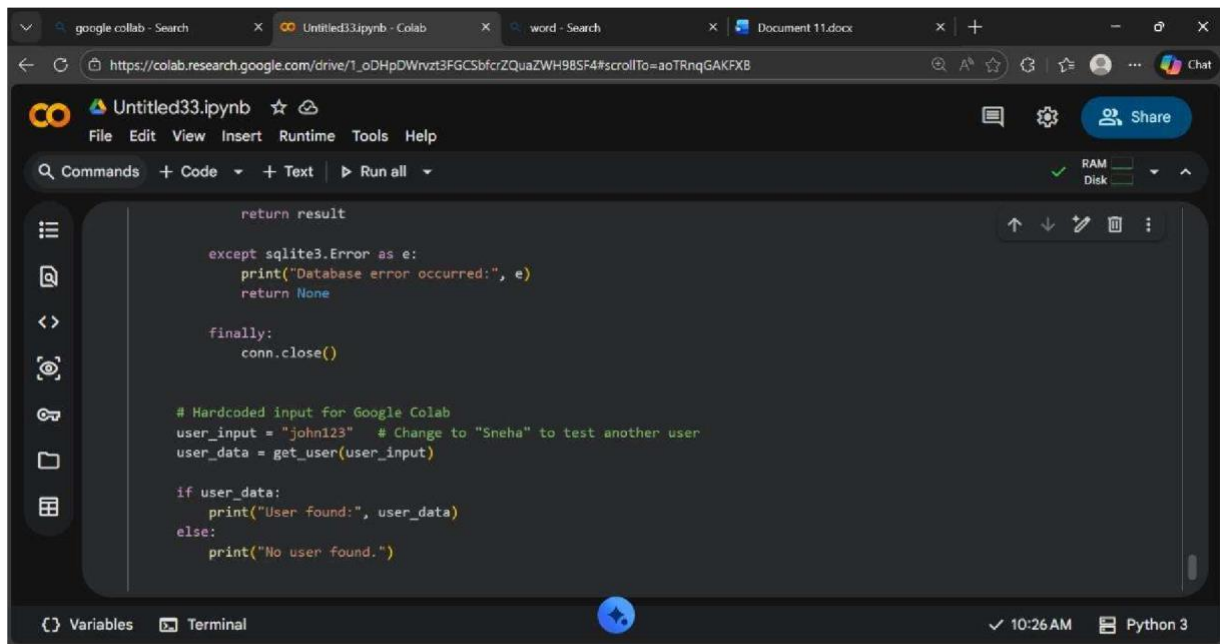
        # Parameterized query prevents SQL injection
        query = "SELECT * FROM users WHERE username = ?"
        cursor.execute(query, (username,))

        result = cursor.fetchone()
        return result

    except sqlite3.Error as e:
        print("Database error occurred:", e)
        return None

    finally:
        conn.close()
```

A 'Snipping Tool' window is overlaid on the bottom right of the code editor, displaying the message: 'Screenshot copied to clipboard. Automatically saved to screenshots folder.' Below this message is a button labeled 'Markup and share'.



```
return result

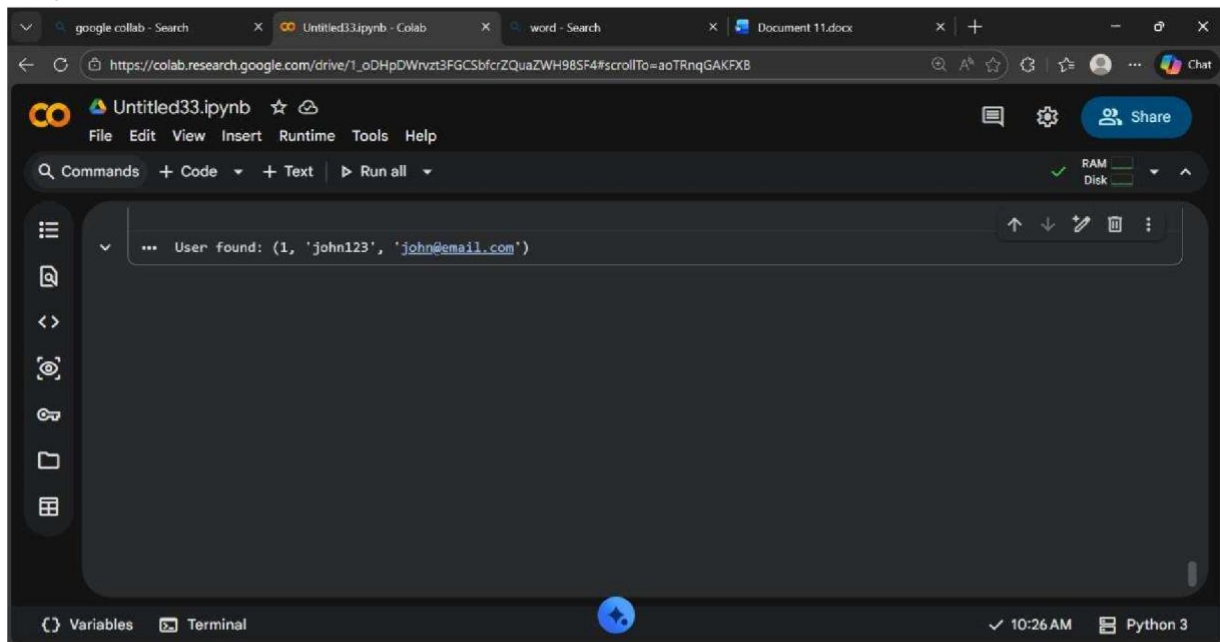
except sqlite3.Error as e:
    print("Database error occurred:", e)
    return None

finally:
    conn.close()

# Hardcoded input for Google Colab
user_input = "john123" # Change to "Sneha" to test another user
user_data = get_user(user_input)

if user_data:
    print("User found:", user_data)
else:
    print("No user found.")
```

Output:



```
... User found: (1, 'john123', 'john@email.com')
```

Explanation:

-->The original code was insecure because it built SQL queries using string concatenation, which could lead to SQL injection attacks.

-->The refactored version uses parameterized queries (?) to safely pass user input to the database.

-->Input validation was added to ensure only alphanumeric usernames are accepted, reducing the risk of malicious input.

-->Try-except blocks were introduced to handle database errors without crashing the program.

--> A finally block ensures the database connection is always closed, improving reliability and making the code safe for production use.

Task 5:

AI-Based Automated Code Review Report

Scenario

Your team uses AI tools to perform automated preliminary code reviews before human review, to improve code quality and consistency across projects.

Task Description

You are provided with a poorly written Python script.

Using AI-assisted review:

- Generate a structured code review report that evaluates:

- o Code readability
- o Naming

- conventions
- o Formatting and style

- consistency
- o Error handling
- o

- Documentation quality
- o

- Maintainability

The task is not just to fix the code, but to analyze and report on quality issues.

Expected Outcome

- AI-generated review report including:

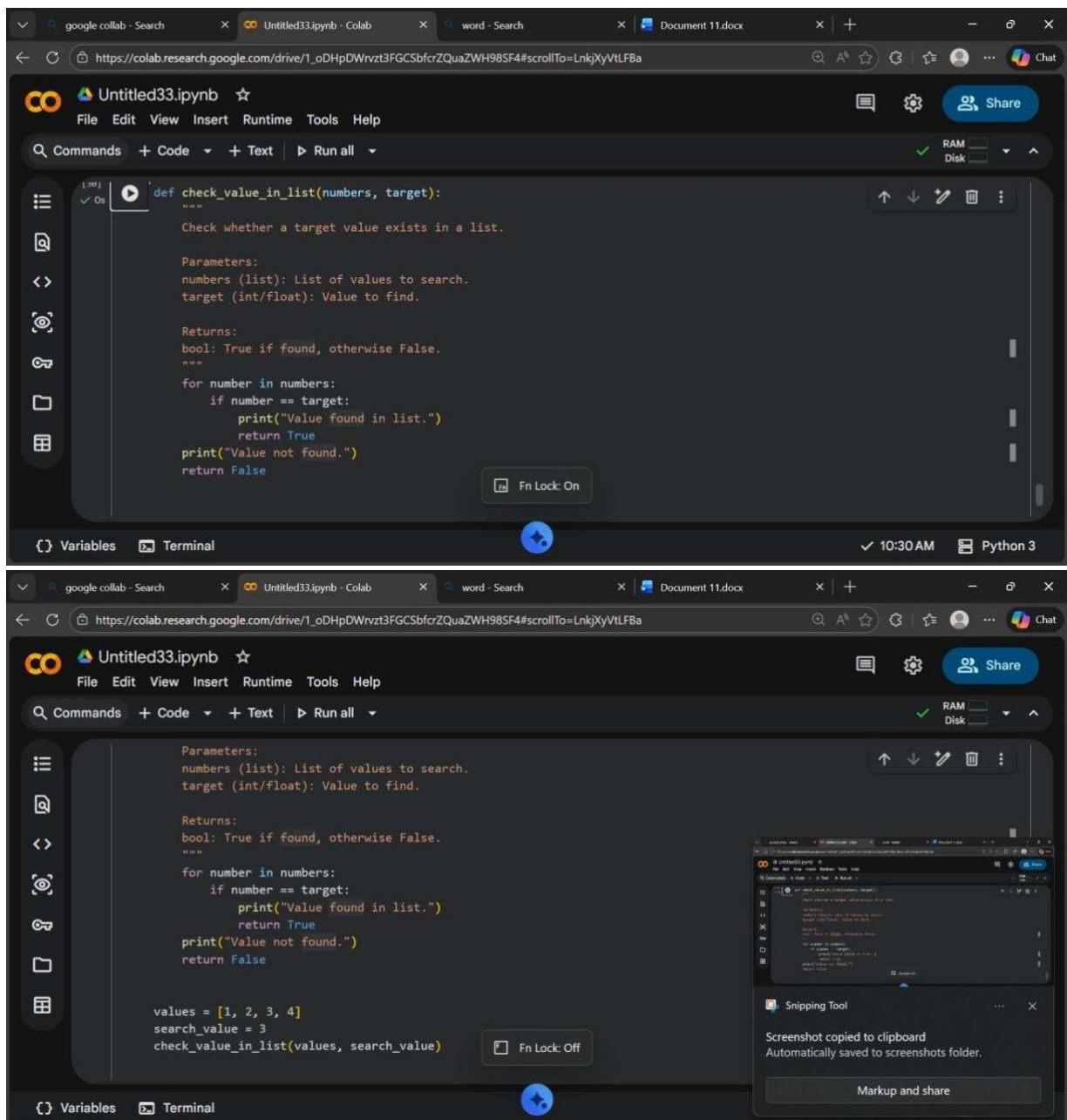
o Identified quality issues o Risk areas o

Code smell detection o Improvement

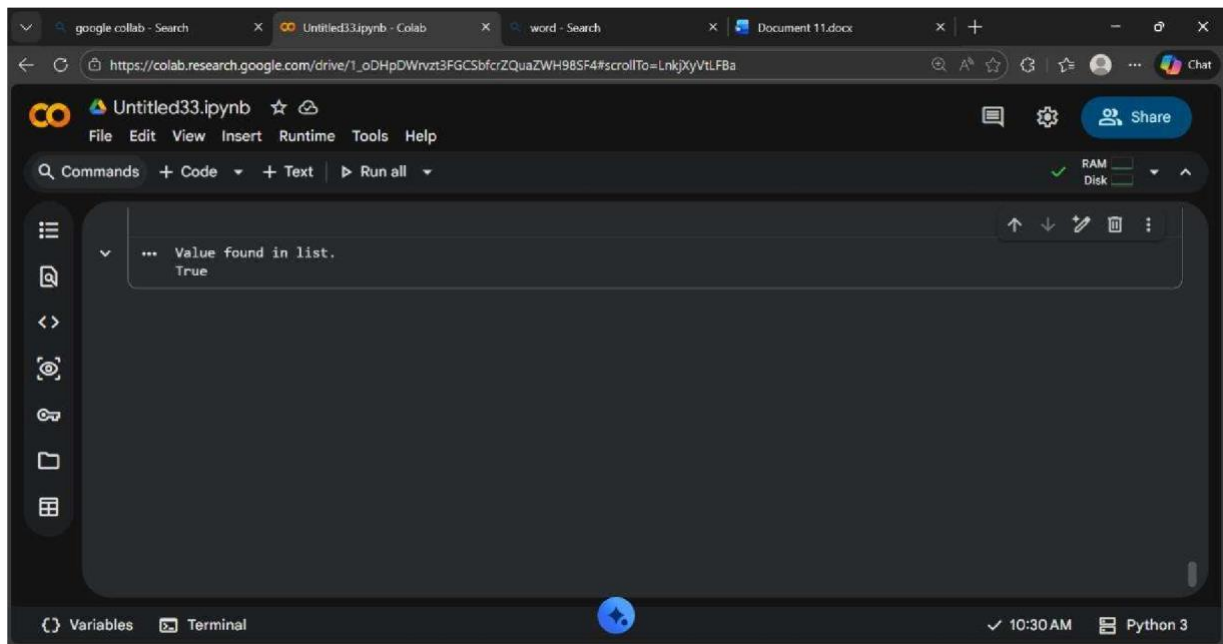
suggestions • Optional improved version

of the code

Code:



Output:



Explanation:

-->In this task, AI was used as a code reviewer to analyze code quality instead of just fixing errors.

-->The AI identified issues related to poor readability, unclear naming, bad formatting, missing documentation, and lack of error handling.

-->It also detected code smells such as unused variables and unnecessary statements. Based on this analysis, improvement suggestions were provided to make the code more maintainable and professional.

-->This demonstrates how AI helps teams perform faster and more consistent preliminary code reviews before human evaluation.